

IMCQ 5 Key
version 1
GIT Physiology
BPM2 module DM

Fall 2020

Dr. Stephan Bandelow

1. A 56-year old obese man comes to the doctor because of a 6-month history of heartburn, difficulty swallowing, gastrointestinal discomfort and a nonspecific feeling of pressure on his chest. He has been taking antacids for almost 20 years and has a 30-year history of smoking. Gastroendoscopy shows ulceration, stricture and columnar cellular morphology with goblet cells. Which of the following is the most likely diagnosis?

- A. Achalasia
- B. Barrett's esophagus
- C. Gastritis
- D. Gastroesophageal reflux
- E. Esophageal Spasm

2. A 70-year old woman comes to the physician because of a 2-week history of tiredness and fatigue. She has recently undergone a gastrectomy because of Zollinger–Ellison syndrome. Physical examination shows no abnormalities. Laboratory studies show megaloblastic macrocytic anemia. She is given a prescription for intramuscular Cobalamin. This deficiency replacement therapy is most likely needed because of the absence of which of the following cell types?

- A. Chief
- B. G
- C. Goblet
- D. Mucous neck
- E. Parietal

3. A 54-year-old man comes to the emergency department because of a 5-month history of nausea and intermittent upper abdominal pain. He does not report any hematemesis. Stool sample studies show some occult blood. Microscopic examination of a gastric biopsy shows acute and chronic mucosal inflammation and *Helicobacter pylori*. Which of the following is the most likely diagnosis?

- A. Atrophic gastritis
- B. Development of pernicious anemia
- C. Esophageal metaplasia
- D. Mucosal invasion with septicemia

4. A 16-year old boy is brought to the physician by his partents because of a 2-week history of difficulty swallowing and regurgitating liquids. Manometry reports confirm absent primary peristaltic contractions of the pharynx and esophagus. Secondary peristalsis of the lower esophagus occurs on distension of the esophageal body. A lesion in which of the following locations is most likely to be associated with these clinical findings?

- A. Cortical region of the brain
- B. Cricopharyngeal muscle
- C. Enteric nerves
- D. Nucleus ambiguus of the vagus
- E. Dorsal motor nucleus

5. A 38-year-old woman comes to the physician because of 5-month history of increasing colicky abdominal pains, primarily in the right upper quadrant. She notes that these pains are particularly pronounced after eating burger and fries. She is in the 6th month of her 3rd pregnancy. Which of the following hormones is most likely associated with the increase in her symptoms after a meal?

- A. Cholecystokinin
- B. Gastrin
- C. Motilin
- D. Pepsin
- E. Secretin

6. A study involving 10 healthy subjects and 25 duodenal ulcer patients is determining gastric responses to cephalic phase vagal stimulation induced by 'sham-feeding' (seeing and smelling food without eating anything). Blood gastrin releasing peptide (GRP) and somatostatin (SST) levels are measured before and after atropinization. During sham-feeding, which of the following changes in GRP and SST levels would most likely occur prior to atropinization?

	GRP	Somatostatin
A.	↑	↓
B.	↑	↑
C.	↓	↓
D.	↓	↑
E.	↔	↓

7. A 42-year-old man comes to the physician because of a 2-day history of epigastric abdominal pain associated with nausea and vomiting. He has a 5-year history of alcoholism. The physician suspects acute pancreatitis. An increase in the blood levels of which of the following enzymes would be most indicative of this diagnosis?

- A. Pepsinogen
- B. Gastric lipase
- C. Pancreatic lipase
- D. Salivary amylase
- E. Intrinsic factor

8. A 23-year-old woman comes to the physician because of a 2-week history of abdominal cramps with episodes of diarrhea and constipation. Her temperature is 37 °C (98.6 °F) and occult stool tests are negative. Which of the following best describes motility in the small intestine in this patient?

- A. Bowel wall distension initiates contractile activity
- B. Contractions occur at the same frequency from duodenum to terminal ileum
- C. Migrating motor complexes occur during digestion
- D. Peristalsis is the only contractile activity during feeding
- E. Vagotomy blocks contractile activity

9. A 49-year-old woman comes to the physician because of a 3-month history of heartburn. Gastric pH measurement shows abnormally high acidity ($\text{pH} = 1.8$), and laboratory studies show elevated serum gastrin levels (180 pg/mL). Her vital signs are all within normal limits, and physical examination is otherwise normal. Which of the following signal transduction pathways is most likely overactive in this patient?

- A. CCK_B receptors linking to cAMP
- B. M_3 receptors linking to cAMP
- C. H_2 receptors linking to cAMP
- D. CCK_B receptors linking to $\text{IP}_3/\text{DAG}/\text{Ca}^{2+}$
- E. M_3 receptors linking to $\text{IP}_3/\text{DAG}/\text{Ca}^{2+}$
- F. H_2 receptors linking to $\text{IP}_3/\text{DAG}/\text{Ca}^{2+}$

10. A 6-year-old girl with cystic fibrosis is brought to the physician because her parents are worried about her poor weight gain and growth. She also passes bulky, greasy stools, and tends to have frequent abdominal pains and constipation. Which of the following is the most likely effect of cystic fibrosis on her pancreatic ductal secretions?

- A. no change
- B. $\uparrow$ Na^+ and H_2O secretion
- C. $\downarrow$ Na^+ and H_2O secretion
- D. $\downarrow$ Na^+ and $\uparrow$ H_2O secretion
- E. $\uparrow$ Na^+ and $\downarrow$ H_2O secretion

11. A 74-year-old woman comes to the physician after being diagnosed with early gastric carcinoma (EGC) in the upper stomach. Proximal gastrectomy of the upper third of the stomach is suggested. Which of the following processes will be most affected by this intervention?

- A. Peristalsis
- B. Basic electrical rhythm
- C. Receptive relaxation
- D. Migrating Motor Complex
- E. Churning and mixing

12. A 52-year-old woman comes to the physician because of a 3-day history of diarrhea. She has only been drinking small amounts because fluids are making her feel nauseous. She receives an oral rehydration solution containing NaCl, glucose, KCl and citrate dissolved in water. Which of the following is the most likely intestinal uptake mechanism being targeted by the rehydration solution?

- A. Sodium-chloride co-transport
- B. Passive sodium absorption via luminal channels
- C. Chloride absorption via CFTR channels
- D. Sodium-glucose co-transport
- E. Active potassium absorption

13. A 58-year-old man comes to the physician because of a 2-month history of esophageal acid reflux. Laboratory studies show that his gastric acid output exceeds normal levels. Gastric acid secretion levels in this patient are most likely to result from a tumor formed by which of the following gastro-intestinal cell types?

- A. S cells
- B. G cells
- C. Chief cells
- D. D cells
- E. Enteroendocrine cells

14. A 1-week-old baby boy is brought to the physician by his mother because of a 5-day history of abdominal distension and not passing any feces. An abdominal x-ray shows a narrowed segment of distal colon and significantly dilated intestines above the obstruction. Which of the following disorders is the most likely cause of his symptoms?

- A. Vagus nerve dysfunction
- B. Irritable bowel syndrome
- C. Reduced pancreatic exocrine secretions
- D. Ileocecal sphincter dysfunction
- E. Impaired ganglionic innervation of the colon

15. An 18-year-old woman comes to the physician because of 4-day history of lower abdominal pain, which starts when she begins eating a meal. The physician suspects colitis. Which of the following GI reflexes is most likely to trigger her colonic pain?

- A. Colocolic
- B. Gastroileal
- C. Rectocolic
- D. Gastrocolic
- E. Duodeonocolic

16. A 65-year-old man comes to the physician because of a 6-month history of heartburn. Gastric pH measurement shows persistently high acid production even after the gastric phase of digestion has ended. An abnormality in enterogastrone signaling from the small intestine is suspected. Which of the following mediators is most likely deficient in this patient?

- A. Somatostatin
- B. Acetylcholine
- C. Secretin
- D. Histamine
- E. Gastrin-releasing peptide

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1. A 46-year old obese man comes to the doctor because of a 6-month history heartburn, difficulty swallowing and a nonspecific feeling of pressure on his chest. He has been taking antacids for the past year. Gastroendoscopy shows some ulceration of the lower esophagus, but an otherwise normal squamous cell epithelium with no columnar cellular morphology. Which of the following is the most likely diagnosis?

- A. Achalasia
- B. Barrett's esophagus
- C. Gastritis
- D. Gastroesophageal reflux
- E. Esophageal Spasm

2. A 70-year old woman comes to the physician because of a 2-week history of tiredness and fatigue. She has recently undergone a gastrectomy because of Zollinger –Ellison syndrome. Physical examination shows no abnormalities. Laboratory studies show megaloblastic macrocytic anemia. She is given a prescription for intramuscular Cobalamin. This deficiency replacement therapy is most likely to compensate for the absence of which of the following substances?

- A. Gastrin
- B. Secretin
- C. Pepsinogen
- D. Gastric lipase
- E. Intrinsic factor

3. A 54-year-old man comes to the emergency department because of a 1-week history of nausea and intermittent upper abdominal pain. He does not report any hematemesis. Stool sample studies show some occult blood. Microscopic examination of a gastric biopsy shows signs of acute and mucosal inflammation, and the presence of bacteria. Which of the following bacteria are most likely found in the sample?

- A. *Clostridium difficile*
- B. *Helicobacter pylori*
- C. *Escherichia coli*
- D. *Staphylococcus aureus*
- E. *Enterobacter cloacae*

4. A 16-year old boy presents to the emergency room after he became concerned with milk regurgitating through his nose at breakfast. He also complains of difficulty swallowing. The doctor notices that that air escapes through the patients nose when he speaks. Manometry reports confirm absent primary peristaltic contractions of the pharynx and esophagus, but present secondary peristalsis of the lower esophagus, occurs on distension of the esophageal body. Which of the following locations would you most likely expect to find a lesion that would be associated with the clinical findings?

1. Cortical region of the brain
2. Cricopharyngeal muscle
3. Enteric nerves
4. Nucleus ambiguus of the vagus
5. Pharyngeal muscle

5. A 38-year-old woman comes to the physician because of 5-month history of increasing colicky abdominal pains, primarily in the right upper quadrant. She notes that these pains occur after most meals, even if they do not contain much fat or protein. She is in the 6th month of her 3rd pregnancy. Which of the following hormones is most likely associated with her pains after a meal?

- A. Cholecystokinin
- B. Gastrin
- C. Secretin
- D. Motilin
- E. Pepsin

6. A study involving 10 healthy subjects and 25 duodenal ulcer patients is determining gastric responses to cephalic phase vagal stimulation induced by 'sham-feeding' (seeing and smelling food without eating anything). Blood gastrin releasing peptide (GRP) and somatostatin (SST) levels are measured before and after atropinization. During sham-feeding, which of the following changes in GRP and SST levels would most likely occur after atropinization?

	GRP	Somatostatin
A.	↑	↓
B.	↑	↑
C.	↓	↓
D.	↓	↑
E.	↑	↔

7. A 42-year-old man comes to the physician because of a 2-day history of abdominal pain associated with nausea and vomiting. He has a 3-year history of peptic ulcers. The physician suspects acute gastritis. An increase in the blood levels of which of the following enzymes would be most indicative of this diagnosis?

- A. Pepsinogen
- B. Pancreatic lipase
- C. Amylase
- D. Trypsinogen
- E. Lactase

8. A 23-year-old woman comes to the physician because of a 2-week history of abdominal cramps with episodes of diarrhoea and constipation. Her temperature is 37 °C (98.6 °F) and occult stool tests are negative. Which of the following best describes motility in the small intestine in this patient?

- A. Luminal distension initiates contractile activity
- B. Contractions occur at the same frequency from duodenum to terminal ileum
- C. Migrating motor complexes occur during digestion
- D. Peristalsis is the only contractile activity during feeding
- E. Vagotomy blocks contractile activity

9. A 49-year-old woman comes to the physician because of a 3-month history of heartburn. Gastric pH measurement shows abnormally high acidity ($\text{pH} = 1.8$), and laboratory studies show elevated serum gastrin levels (180 pg/mL). Her vital signs are all within normal limits, and physical examination shows no abnormalities. Which of the following signal transduction pathways is most likely overactive in this patient?

- A. CCK_B receptors linking to $\text{IP}_3/\text{DAG}/\text{Ca}^{2+}$
- B. M_3 receptors linking to $\text{IP}_3/\text{DAG}/\text{Ca}^{2+}$
- C. H_2 receptors linking to $\text{IP}_3/\text{DAG}/\text{Ca}^{2+}$
- D. CCK_B receptors linking to cAMP
- E. M_3 receptors linking to cAMP
- F. H_2 receptors linking to cAMP

10. A 6-year-old girl with cystic fibrosis is brought to the physician because her parents are worried about her poor weight gain and growth. She also passes bulky, greasy stools, and tends to have frequent abdominal pains and constipation. Which of the following is the most likely effect of cystic fibrosis on her pancreatic ductal secretions?

- A. no change
- B. ↓ Cl⁻ and H₂O secretion
- C. ↑ Cl⁻ and H₂O secretion
- D. ↓ Cl⁻ and ↑ H₂O secretion
- E. ↑ Cl⁻ and ↓ H₂O secretion

11. A 74-year-old woman comes to the physician after being diagnosed with early gastric carcinoma (EGC) in the lower stomach. Proximal gastrectomy of the lower third of the stomach is suggested. Which of the following processes would be most affected by this intervention?

- A. Peristalsis
- B. Basic electrical rhythm
- C. Receptive relaxation
- D. Migrating Motor Complex
- E. Retropulsion

12. A 52-year-old woman comes to the physician because of a 3-day history of diarrhea. She has only been drinking small amounts because fluids are making her feel nauseous. She receives an oral rehydration solution containing NaCl, glucose, KCl and citrate dissolved in water. Which of the following is the most likely intestinal uptake mechanism being targeted by the rehydration solution?

- A. Sodium-chloride co-transport
- B. Sodium-glucose co-transport
- C. Chloride absorption via CFTR channels
- D. Passive sodium absorption via eNaC channels
- E. Active potassium absorption

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- A. Sodium-chloride co-transport
- B. Sodium-glucose co-transport
- C. Chloride absorption via CFTR channels
- D. Passive sodium absorption via eNaC channels
- E. Active potassium absorption

13. A 58-year-old man comes to the physician because of a 2-month history of esophageal acid reflux. Laboratory studies show that his gastric acid output exceeds normal levels. Gastric acid secretion levels in this patient are most likely to result from a tumor formed by which of the following gastrointestinal cell types?

- A. ECL cells
- B. S cells
- C. Chief cells
- D. D cells
- E. Enteroendocrine cells

14. A 1-week-old baby boy is brought to the physician because of a 2-day history of abdominal distension and not passing any feces for 5 days. An abdominal x-ray shows a narrowed segment of distal colon and significantly dilated intestines above the obstruction. Which of the following disorders is the most likely diagnosis?

- A. Vagus nerve dysfunction
- B. Irritable bowel syndrome
- C. Reduced pancreatic exocrine secretions
- D. Toxic megacolon
- E. Lactose intolerance

15. An 18-year-old woman comes to the physician because of 4-day history of lower abdominal pain when passing feces. There are no signs of constipation and the physician suspects colitis. Which of the following GI reflexes is most likely to trigger her colonic pain?

- A. Colocolic
- B. Gastroileal
- C. Gastrocolic
- D. Duodenocolic
- E. Rectocolic

16. A 65-year-old man comes to the physician because of a 6-month history of heartburn. Gastric pH measurement shows persistently high acid production even after the gastric phase of digestion has ended. An abnormality in enterogastrone signaling from the small intestine is suspected. Which of the following mediators is most likely deficient in this patient?

- A. Somatostatin
- B. Cholecystikinin
- C. Acetylcholine
- D. Gastrin
- E. Gastrin-releasing peptide

Answer Key

IMCQ 5 version 1

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|------|-------|
| 1. B | 9. D |
| 2. E | 10. C |
| 3. A | 11. C |
| 4. D | 12. D |
| 5. A | 13. B |
| 6. A | 14. E |
| 7. C | 15. D |
| 8. A | 16. C |

Answer Key

IMCQ 5 version 2

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|------|-------|
| 1. D | 9. D |
| 2. E | 10. B |
| 3. B | 11. A |
| 4. D | 12. B |
| 5. C | 13. A |
| 6. E | 14. D |
| 7. A | 15. E |
| 8. A | 16. B |